

Tutorial: Custom YOLO Object Detection with RoboFlow

In this tutorial, you'll learn how to create a custom YOLO object detection model using your own classes and annotations with RoboFlow.

Step 1: Install Required Dependencies

```
In [2]: %pip install --upgrade ultralytics
        %pip install opencv-python
        %pip install roboflow
        %pip install matplotlib
        %pip install numpy==1.26.4 --force-reinstall
```

Requirement already satisfied: ultralytics in c:\users\brand\anaconda3\envs\dev\lib\site-packages (8.4.26)

Requirement already satisfied: numpy>=1.23.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (2.1.3)

Requirement already satisfied: matplotlib>=3.3.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (3.10.8)

Requirement already satisfied: opencv-python>=4.6.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (4.13.0.92)

Requirement already satisfied: pillow>=7.1.2 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (12.0.0)

Requirement already satisfied: pyyaml>=5.3.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (6.0.3)

Requirement already satisfied: requests>=2.23.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (2.32.5)

Requirement already satisfied: scipy>=1.4.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (1.17.0)

Collecting torch>=1.8.0 (from ultralytics)

Using cached torch-2.11.0-cp313-cp313-win_amd64.whl.metadata (29 kB)

Collecting torchvision>=0.9.0 (from ultralytics)

Using cached torchvision-0.26.0-cp313-cp313-win_amd64.whl.metadata (5.5 kB)

Requirement already satisfied: psutil>=5.8.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (7.2.2)

Requirement already satisfied: polars>=0.20.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (1.39.3)

Requirement already satisfied: ultralytics-thop>=2.0.18 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from ultralytics) (2.0.18)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib>=3.3.0->ultralytics) (1.3.3)

Requirement already satisfied: cycler>=0.10 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib>=3.3.0->ultralytics) (4.61.0)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib>=3.3.0->ultralytics) (1.4.9)

Requirement already satisfied: packaging>=20.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib>=3.3.0->ultralytics) (25.0)

Requirement already satisfied: pyparsing>=3 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib>=3.3.0->ultralytics) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)

Requirement already satisfied: polars-runtime-32==1.39.3 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from polars>=0.20.0->ultralytics) (1.39.3)

Requirement already satisfied: six>=1.5 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from requests>=2.23.0->ultralytics) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from requests>=2.23.0->ultralytics) (3.7)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from requests>=2.23.0->ultralytics) (2.6.3)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from requests>=2.23.0->ultralytics) (2026.1.4)

Requirement already satisfied: filelock in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from torch>=1.8.0->ultralytics) (3.25.2)

Requirement already satisfied: typing-extensions>=4.10.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from torch>=1.8.0->ultralytics) (4.15.0)

Requirement already satisfied: setuptools<82 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from torch>=1.8.0->ultralytics) (80.10.1)

Requirement already satisfied: sympy>=1.13.3 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from torch>=1.8.0->ultralytics) (1.13.3)

[illegible]

```

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----- 2/2 [torchvision]

```

Successfully installed torch-2.11.0 torchvision-0.26.0

Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: opencv-python in c:\users\brand\anaconda3\envs\dev\lib\site-packages (4.13.0.92)

Requirement already satisfied: numpy>=2 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from opencv-python) (2.1.3)

Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: roboflow in c:\users\brand\anaconda3\envs\dev\lib\site-packages (1.2.16)

Requirement already satisfied: certifi in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (2026.1.4)

Requirement already satisfied: idna==3.7 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (3.7)

Requirement already satisfied: cyciler in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (0.12.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (1.4.9)

Requirement already satisfied: matplotlib in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (3.10.8)

Requirement already satisfied: numpy>=1.18.5 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (2.1.3)

Requirement already satisfied: opencv-python-headless==4.10.0.84 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (4.10.0.84)

Requirement already satisfied: Pillow>=7.1.2 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (12.0.0)

Requirement already satisfied: pillow-avif-plugin<2 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (1.5.5)

Requirement already satisfied: python-dateutil in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (2.9.0)

ev\lib\site-packages (from roboflow) (2.9.0.post0)
Requirement already satisfied: python-dotenv in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (1.2.2)
Requirement already satisfied: requests in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (2.32.5)
Requirement already satisfied: six in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (1.17.0)
Requirement already satisfied: urllib3>=1.26.6 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (2.6.3)
Requirement already satisfied: tqdm>=4.41.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (4.67.3)
Requirement already satisfied: PyYAML>=5.3.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (6.0.3)
Requirement already satisfied: requests-toolbelt in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (1.0.0)
Requirement already satisfied: filetype in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (1.2.0)
Requirement already satisfied: pi-heif<2 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from roboflow) (1.3.0)
Requirement already satisfied: colorama in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from tqdm>=4.41.0->roboflow) (0.4.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib->roboflow) (1.3.3)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib->roboflow) (4.61.0)
Requirement already satisfied: packaging>=20.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib->roboflow) (25.0)
Requirement already satisfied: pyparsing>=3 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib->roboflow) (3.2.5)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from requests->roboflow) (3.4.4)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: matplotlib in c:\users\brand\anaconda3\envs\dev\lib\site-packages (3.10.8)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (4.61.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (2.1.3)
Requirement already satisfied: packaging>=20.0 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=3 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\brand\anaconda3\envs\dev\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
Collecting numpy==1.26.4
Using cached numpy-1.26.4.tar.gz (15.8 MB)
Installing build dependencies: started
Installing build dependencies: finished with status 'done'
Getting requirements to build wheel: started

```
Getting requirements to build wheel: finished with status 'done'  
Installing backend dependencies: started  
Installing backend dependencies: finished with status 'done'  
Preparing metadata (pyproject.toml): started  
Preparing metadata (pyproject.toml): finished with status 'error'  
Note: you may need to restart the kernel to use updated packages.
```

error: subprocess-exited-with-error

× Preparing metadata (pyproject.toml) did not run successfully.

| exit code: 1

└> [21 lines of output]

```
+ c:\Users\brand\anaconda3\envs\dev\python.exe C:\Users\brand\AppData\Local\Temp\pip-install-pd_3xf4k\numpy_096e098a68654d6e9d47b846c987ed4e\vendored-meson\meson.py setup C:\Users\brand\AppData\Local\Temp\pip-install-pd_3xf4k\numpy_096e098a68654d6e9d47b846c987ed4e C:\Users\brand\AppData\Local\Temp\pip-install-pd_3xf4k\numpy_096e098a68654d6e9d47b846c987ed4e\mesonpy-oe0tfsx6 -Dbuildtype=release -Db_ndebug=if-release -Db_vsctrl=md --native-file=C:\Users\brand\AppData\Local\Temp\pip-install-pd_3xf4k\numpy_096e098a68654d6e9d47b846c987ed4e\mesonpy-oe0tfsx6\meson-python-native-file.ini
```

```
The Meson build system
```

```
Version: 1.2.99
```

```
Source dir: C:\Users\brand\AppData\Local\Temp\pip-install-pd_3xf4k\numpy_096e098a68654d6e9d47b846c987ed4e
```

```
Build dir: C:\Users\brand\AppData\Local\Temp\pip-install-pd_3xf4k\numpy_096e098a68654d6e9d47b846c987ed4e\mesonpy-oe0tfsx6
```

```
Build type: native build
```

```
Project name: NumPy
```

```
Project version: 1.26.4
```

```
WARNING: Failed to activate VS environment: Could not find C:\Program Files (x86)\Microsoft Visual Studio\Installer\vswhere.exe
```

```
..\meson.build:1:0: ERROR: Unknown compiler(s): [['icl'], ['cl'], ['cc'], ['gcc'], ['clang'], ['clang-cl'], ['pgcc']]
```

```
The following exception(s) were encountered:
```

```
Running `icl ""` gave "[WinError 2] El sistema no puede encontrar el archivo especificado"
```

```
Running `cl /?` gave "[WinError 2] El sistema no puede encontrar el archivo especificado"
```

```
Running `cc --version` gave "[WinError 2] El sistema no puede encontrar el archivo especificado"
```

```
Running `gcc --version` gave "[WinError 2] El sistema no puede encontrar el archivo especificado"
```

```
Running `clang --version` gave "[WinError 2] El sistema no puede encontrar el archivo especificado"
```

```
Running `clang-cl /?` gave "[WinError 2] El sistema no puede encontrar el archivo especificado"
```

```
Running `pgcc --version` gave "[WinError 2] El sistema no puede encontrar el archivo especificado"
```

```
A full log can be found at C:\Users\brand\AppData\Local\Temp\pip-install-pd_3xf4k\numpy_096e098a68654d6e9d47b846c987ed4e\mesonpy-oe0tfsx6\meson-logs\meson-log.txt
```

```
[end of output]
```

note: This error originates from a subprocess, and is likely not a problem with pip.

error: metadata-generation-failed

× Encountered error while generating package metadata.

└> numpy

note: This is an issue with the package mentioned above, not pip.

hint: See above for details.

Step 2: Import Required Libraries

```
In [ ]: import os
os.environ["KMP_DUPLICATE_LIB_OK"] = "TRUE" # Necesario para evitar que muera el
import cv2
from roboflow import Roboflow
from ultralytics import YOLO
import matplotlib.pyplot as plt
import numpy as np
from IPython.display import Image, display
```

Step 3: Set Up RoboFlow Account

1. Go to RoboFlow and create a free account
2. Create a new project and select "Object Detection" as project type
3. Choose "YOLOv26" as the model format

Step 4: Upload and Annotate Your Images

Using RoboFlow Web Interface:

1. Click "Upload" to add your images
2. Use the annotation tool to draw bounding boxes around objects
3. Assign the correct class labels to each bounding box
4. Apply preprocessing and augmentation as needed

Step 5. Download your Custom your Dataset

1. In RoboFlow, go to your project dashboard
2. Click "Generate" to create a dataset version
3. Apply any desired augmentations
4. Click "Export" and select "YOLOv~~x~~" format (~~x~~ the version you are using)
5. Copy the download code

```
In [ ]: # Initialize RoboFlow (you'll need your API key)

from roboflow import Roboflow
rf = Roboflow(api_key="eGFYeFonZgeWFCSZdR7E")
project = rf.workspace("brandons-workspace-dklql").project("tareasemana3")
dataset = project.version(3).download("yolo26") # Se crearon multiples versiones
```

```
loading Roboflow workspace...
loading Roboflow project...
```

Step 6: Train Your Custom YOLO Model

```
In [ ]: # Load the YOLO model with custom configuration
model = YOLO('yolo26n.pt') # Using nano version as base
```

```
# Train the model on your custom dataset
results = model.train(
    data=f'{dataset.location}/data.yaml', # Path to dataset config
    epochs=50, # Se aumento de 20 -> 50 para mejorar el entrenamiento, este fue
    imgsz=640, # Se mantuvo la calidad de imagen para evitar perdidas de calida
    batch=8, # Se paso de 16 -> 8 para evitar que se consuma demasiada memoria
    name='yolo_clase_3', # Name for the training run
    device='cuda' # Se agrego para forzar el uso de La GPU en lugar de La CPU
)
```

Ultralytics 8.4.26 Python-3.13.11 torch-2.11.0+cu126 CUDA:0 (NVIDIA GeForce RTX 4060 Ti, 8188MiB)

engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=c:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\TareaSemana3-3\data.yaml, degrees=0.0, deterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=50, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo26n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=yolo_clase_3, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\runs\detect\yolo_clase_3, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None

Overriding model.yaml nc=80 with nc=3

	from	n	params	module
arguments				
0	-1	1	464	ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]				
1	-1	1	4672	ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]				
2	-1	1	6640	ultralytics.nn.modules.block.C3k2
[32, 64, 1, False, 0.25]				
3	-1	1	36992	ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]				
4	-1	1	26080	ultralytics.nn.modules.block.C3k2
[64, 128, 1, False, 0.25]				
5	-1	1	147712	ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]				
6	-1	1	87040	ultralytics.nn.modules.block.C3k2
[128, 128, 1, True]				
7	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
8	-1	1	346112	ultralytics.nn.modules.block.C3k2
[256, 256, 1, True]				
9	-1	1	164608	ultralytics.nn.modules.block.SPPF
[256, 256, 5, 3, True]				
10	-1	1	249728	ultralytics.nn.modules.block.C2PSA
[256, 256, 1]				
11	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat
[1]				
13	-1	1	119808	ultralytics.nn.modules.block.C3k2
[384, 128, 1, True]				
14	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				
15	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat
[1]				

```

16          -1  1    34304  ultralytics.nn.modules.block.C3k2
[256, 64, 1, True]
17          -1  1    36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
18          [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat
[1]
19          -1  1    95232  ultralytics.nn.modules.block.C3k2
[192, 128, 1, True]
20          -1  1   147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
21          [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat
[1]
22          -1  1   463104  ultralytics.nn.modules.block.C3k2
[384, 256, 1, True, 0.5, True]
23          [16, 19, 22]  1   242346  ultralytics.nn.modules.head.Detect
[3, 1, True, [64, 128, 256]]
YOLO26n summary: 260 layers, 2,504,970 parameters, 2,504,970 gradients, 5.8 GFLOPs

```

Transferred 606/708 items from pretrained weights

AMP: running Automatic Mixed Precision (AMP) checks...

AMP: checks passed

train: Fast image access (ping: 0.20.1 ms, read: 96.730.1 MB/s, size: 24.4 KB)

train: Scanning C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\TareaSemana3-3\train\labels.cache... 58 images, 0 backgrounds, 0 corrupt: 100% ————— 58/58 17.4Mit/s 0.0s

val: Fast image access (ping: 0.20.1 ms, read: 66.46.3 MB/s, size: 22.8 KB)

val: Scanning C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\TareaSemana3-3\valid\labels.cache... 25 images, 0 backgrounds, 0 corrupt: 100% ————— 25/25 6.2Mit/s 0.0s

optimizer: 'optimizer=auto' found, ignoring 'lr=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr' and 'momentum' automatically...

optimizer: AdamW(lr=0.001429, momentum=0.9) with parameter groups 114 weight(decay=0.0), 126 weight(decay=0.0005), 126 bias(decay=0.0)

Plotting labels to C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\runs\detect\yolo_clase_3\labels.jpg...

Image sizes 640 train, 640 val

Using 8 dataloader workers

Logging results to C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\runs\detect\yolo_clase_3

Starting training for 50 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/50	1.25G	1.176	7.572	0.02521	3	640: 10
0% —————	8/8 3.6s/it	28.5s	0.2s			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 1.8s/it	3.5s	6.1s			
	all	25	25	0.00605	0.889	0.0476
0.0336						

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
2/50	1.42G	0.9468	7.218	0.02037	4	640: 10
0% —————	8/8 10.2it/s	0.8s	1s			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 15.0it/s	0.1s				
	all	25	25	0.00448	1	0.26
0.174						

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
3/50	1.42G	0.8737	6.039	0.01553	5	640: 10

0%	8/8 10.4it/s 0.8s.1s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 18.2it/s 0.1s						
	all	25	25	0.00388	1	0.376	
0.28							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
4/50	1.42G	0.8059	6.111	0.01564	4	640: 10	
0%	8/8 10.9it/s 0.7s.3s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 20.1it/s 0.1s						
	all	25	25	0.00349	1	0.487	
0.363							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
5/50	1.42G	0.7661	5.982	0.01295	8	640: 10	
0%	8/8 11.3it/s 0.7s.1s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 19.0it/s 0.1s						
	all	25	25	0.00329	0.972	0.486	
0.429							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
6/50	1.42G	0.8021	6.046	0.01531	5	640: 10	
0%	8/8 12.2it/s 0.7s.1s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 19.3it/s 0.1s						
	all	25	25	0.00344	1	0.492	
0.409							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
7/50	1.42G	0.8048	5.78	0.01723	4	640: 10	
0%	8/8 11.9it/s 0.7s.3s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 20.0it/s 0.1s						
	all	25	25	0.00344	1	0.476	
0.431							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
8/50	1.42G	0.8315	5.87	0.01665	2	640: 10	
0%	8/8 12.4it/s 0.6s.1s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 19.7it/s 0.1s						
	all	25	25	0.00341	1	0.528	
0.465							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
9/50	1.42G	0.8515	5.89	0.0189	2	640: 10	
0%	8/8 12.1it/s 0.7s.3s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 19.6it/s 0.1s						
	all	25	25	0.00348	1	0.549	
0.499							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
10/50	1.42G	0.9133	5.89	0.01855	3	640: 10	
0%	8/8 12.2it/s 0.7s.2s						
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2 19.3it/s 0.1s						
	all	25	25	0.634	0.293	0.619	

0.548

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	11/50	1.42G	0.8862	5.159	0.01811	5	640: 10
0%	8/8 11.9it/s 0.7s.1s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.3it/s 0.1s						
	all		25	25	0.664	0.263	0.645

0.543

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	12/50	1.42G	0.8106	5.09	0.0179	3	640: 10
0%	8/8 12.4it/s 0.6s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 19.8it/s 0.1s						
	all		25	25	0.81	0.23	0.623

0.535

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	13/50	1.42G	0.8895	5.255	0.01954	5	640: 10
0%	8/8 12.5it/s 0.6s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.1it/s 0.1s						
	all		25	25	0.683	0.426	0.709

0.614

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	14/50	1.42G	0.8697	5.113	0.01896	3	640: 10
0%	8/8 12.6it/s 0.6s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.0it/s 0.1s						
	all		25	25	0.654	0.675	0.766

0.655

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	15/50	1.42G	0.9252	5.064	0.01842	3	640: 10
0%	8/8 12.5it/s 0.6s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.9it/s 0.1s						
	all		25	25	0.749	0.639	0.804

0.713

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	16/50	1.42G	0.8014	4.839	0.01477	3	640: 10
0%	8/8 12.6it/s 0.6s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.0it/s 0.1s						
	all		25	25	0.737	0.631	0.813

0.732

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	17/50	1.42G	0.8503	4.657	0.01773	8	640: 10
0%	8/8 12.6it/s 0.6s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.2it/s 0.1s						
	all		25	25	0.734	0.537	0.79

0.721

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	18/50	1.42G	0.8262	4.205	0.0173	5	640: 10

0%	8/8	12.4it/s	0.6s.2s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	19.8it/s	0.1s				
	all	25	25	0.734	0.654	0.807	
0.729							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
19/50	1.42G	0.8774	4.75	0.018	4	640: 10	
0%	8/8	10.5it/s	0.8s.3s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	18.9it/s	0.1s				
	all	25	25	0.669	0.661	0.821	
0.733							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
20/50	1.42G	0.8419	4.009	0.0154	6	640: 10	
0%	8/8	12.0it/s	0.7s.2s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	20.2it/s	0.1s				
	all	25	25	0.691	0.782	0.867	
0.794							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
21/50	1.42G	0.8356	4.069	0.01675	5	640: 10	
0%	8/8	11.6it/s	0.7s.2s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	19.4it/s	0.1s				
	all	25	25	0.895	0.825	0.908	
0.819							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
22/50	1.42G	0.749	3.907	0.01449	8	640: 10	
0%	8/8	11.3it/s	0.7s.3s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	18.2it/s	0.1s				
	all	25	25	0.843	0.791	0.92	
0.836							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
23/50	1.42G	0.8696	4.224	0.01685	5	640: 10	
0%	8/8	12.2it/s	0.7s.2s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	18.6it/s	0.1s				
	all	25	25	0.877	0.838	0.925	
0.838							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
24/50	1.42G	0.6426	4.131	0.01294	2	640: 10	
0%	8/8	12.9it/s	0.6s.2s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	21.1it/s	0.1s				
	all	25	25	0.665	0.907	0.877	
0.757							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
25/50	1.42G	0.8599	4.045	0.01808	5	640: 10	
0%	8/8	12.8it/s	0.6s.2s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	2/2	20.7it/s	0.1s				
	all	25	25	0.776	0.88	0.879	

0.777

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	26/50	1.42G	0.756	3.63	0.01451	8	640: 10
0%	8/8 12.4it/s 0.6s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.0it/s 0.1s						
	all		25	25	0.797	0.865	0.884

0.807

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	27/50	1.42G	0.7329	3.918	0.01436	5	640: 10
0%	8/8 12.2it/s 0.7s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 19.8it/s 0.1s						
	all		25	25	0.76	0.892	0.874

0.77

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	28/50	1.42G	0.8166	3.994	0.01834	2	640: 10
0%	8/8 10.6it/s 0.8s.1s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 19.7it/s 0.1s						
	all		25	25	0.772	0.918	0.922

0.8

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	29/50	1.42G	0.8149	3.628	0.01635	3	640: 10
0%	8/8 12.1it/s 0.7s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 19.6it/s 0.1s						
	all		25	25	0.804	0.892	0.959

0.838

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	30/50	1.42G	0.8234	3.422	0.01474	5	640: 10
0%	8/8 10.7it/s 0.7s.3s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 19.8it/s 0.1s						
	all		25	25	0.942	0.753	0.95

0.829

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	31/50	1.42G	0.7912	3.45	0.01675	3	640: 10
0%	8/8 11.5it/s 0.7s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 18.8it/s 0.1s						
	all		25	25	0.849	0.775	0.923

0.813

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	32/50	1.42G	0.7927	4.129	0.01656	2	640: 10
0%	8/8 12.1it/s 0.7s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.2it/s 0.1s						
	all		25	25	0.668	0.876	0.915

0.825

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	33/50	1.42G	0.7402	3.128	0.01425	5	640: 10

0%	8/8	12.2it/s	0.7s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	18.4it/s	0.1s				
	all	25	25	0.838	0.778	0.94	
0.866							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
34/50	1.42G	0.751	3.238	0.01442	5	640: 10	
0%	8/8	12.1it/s	0.7s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	20.1it/s	0.1s				
	all	25	25	0.785	0.875	0.936	
0.841							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
35/50	1.42G	0.9346	3.865	0.01957	5	640: 10	
0%	8/8	12.4it/s	0.6s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	20.1it/s	0.1s				
	all	25	25	0.875	0.793	0.944	
0.841							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
36/50	1.42G	0.7186	3.202	0.01344	5	640: 10	
0%	8/8	12.7it/s	0.6s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	19.1it/s	0.1s				
	all	25	25	0.874	0.81	0.936	
0.84							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
37/50	1.42G	0.7357	3.561	0.01483	5	640: 10	
0%	8/8	12.4it/s	0.6s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	20.1it/s	0.1s				
	all	25	25	0.749	0.942	0.944	
0.857							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
38/50	1.42G	0.7139	3.081	0.01359	6	640: 10	
0%	8/8	12.2it/s	0.7s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	18.7it/s	0.1s				
	all	25	25	0.758	0.955	0.946	
0.861							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
39/50	1.42G	0.8228	3.022	0.01561	5	640: 10	
0%	8/8	12.4it/s	0.6s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	19.9it/s	0.1s				
	all	25	25	0.882	0.803	0.946	
0.865							

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
40/50	1.42G	0.7729	3.449	0.01579	3	640: 10	
0%	8/8	12.5it/s	0.6s.2s				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	2/2	20.0it/s	0.1s				
	all	25	25	0.922	0.814	0.948	

0.868

Closing dataloader mosaic

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	41/50	1.42G	0.5303	5.257	0.01674	2	640: 10
0%	8/8 5.2it/s 1.5s0.1s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.2it/s 0.1s						
	all		25	25	0.885	0.824	0.952

0.873

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	42/50	1.42G	0.5121	5.188	0.01499	2	640: 10
0%	8/8 12.1it/s 0.7s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 19.6it/s 0.1s						
	all		25	25	0.932	0.852	0.954

0.866

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	43/50	1.42G	0.4765	5.093	0.01376	2	640: 10
0%	8/8 12.0it/s 0.7s.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 18.8it/s 0.1s						
	all		25	25	0.939	0.849	0.949

0.86

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	44/50	1.42G	0.4547	5.067	0.01309	2	640: 10
0%	8/8 11.6it/s 0.7s.3s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 19.0it/s 0.1s						
	all		25	25	0.951	0.89	0.948

0.859

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	45/50	1.42G	0.4757	5.027	0.01507	2	640: 10
0%	8/8 11.3it/s 0.7s.3s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 20.1it/s 0.1s						
	all		25	25	0.875	0.88	0.924

0.843

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	46/50	1.42G	0.5017	5.132	0.01528	2	640: 10
0%	8/8 7.2it/s 1.1s0.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 9.5it/s 0.2s0.3s						
	all		25	25	0.872	0.833	0.933

0.854

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	47/50	1.42G	0.5039	5.13	0.01575	2	640: 10
0%	8/8 4.9it/s 1.6s0.2s						
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	2/2 6.7it/s 0.3s0.5s						
	all		25	25	0.842	0.934	0.956

0.864

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
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48/50	1.42G	0.4928	5.023	0.01484	2	640: 10
0% —————	8/8 5.1it/s	1.6s0.2s				
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100% —————	2/2	8.7it/s	0.2s0.4s			
	all	25	25	0.843	0.927	0.949

0.851

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
49/50	1.42G	0.4359	5.065	0.01355	2	640: 10
0% —————	8/8 5.6it/s	1.4s0.2s				
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100% —————	2/2	8.3it/s	0.2s0.4s			
	all	25	25	0.886	0.872	0.942

0.845

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
50/50	1.42G	0.4898	5.024	0.01519	2	640: 10
0% —————	8/8 5.7it/s	1.4s0.3s				
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100% —————	2/2	15.3it/s	0.1s			
	all	25	25	0.837	0.939	0.946

0.849

50 epochs completed in 0.027 hours.

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Optimizer stripped from C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\runs\detect\yolo_clase_3\weights\best.pt, 5.4MB

Validating C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\runs\detect\yolo_clase_3\weights\best.pt...

Ultralytics 8.4.26 Python-3.13.11 torch-2.11.0+cu126 CUDA:0 (NVIDIA GeForce RTX 4060 Ti, 8188MiB)

YOLO26n summary (fused): 122 layers, 2,375,421 parameters, 0 gradients, 5.2 GFLOPs

	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100% —————	2/2	12.1it/s	0.2s.3s			
	all	25	25	0.883	0.824	0.952
0.872						
	Dinosaurio	12	12	0.936	0.833	0.968
0.895						
	Llaves	9	9	0.835	0.889	0.975
0.892						
	Pin	4	4	0.879	0.75	0.912
0.829						

Speed: 0.4ms preprocess, 2.8ms inference, 0.0ms loss, 0.1ms postprocess per image

Results saved to C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_clase3\runs\detect\yolo_clase_3

Step 7: Evaluate Your Model

```
In [4]: # Evaluate model performance
metrics = model.val() # Validate the model
print(f"mAP50-95: {metrics.box.map}")
print(f"mAP50: {metrics.box.map50}")

# Show some validation results
!yolo task=detect mode=val model=runs/detect/yolo_clase_3/weights/best.pt data={
```

```

Ultralytics 8.4.26 Python-3.13.11 torch-2.11.0+cu126 CUDA:0 (NVIDIA GeForce RTX
4060 Ti, 8188MiB)
YOLO26n summary (fused): 122 layers, 2,375,421 parameters, 0 gradients, 5.2 GFLOP
s
val: Fast image access (ping: 0.00.0 ms, read: 492.484.8 MB/s, size: 21.6 KB)
val: Scanning C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_cla
se3\TareaSemana3-3\valid\labels.cache... 25 images, 0 backgrounds, 0 corrupt: 10
0% ————— 25/25 9.5Mit/s 0.0s

```

	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	all	2/2	1.4s/it 2.7s8.5s	0.883	0.824	0.952	
0.87	Dinosaurio	12	12	0.937	0.833	0.968	
0.889	Llaves	9	9	0.836	0.889	0.975	
0.892	Pin	4	4	0.877	0.75	0.912	
0.829							

```

Speed: 6.0ms preprocess, 14.5ms inference, 0.0ms loss, 0.2ms postprocess per imag
e
Results saved to C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_
clase3\runs\detect\val
mAP50-95: 0.8698585931949435
mAP50: 0.9516666666666667
Ultralytics 8.4.26 Python-3.13.11 torch-2.11.0+cu126 CUDA:0 (NVIDIA GeForce RTX
4060 Ti, 8188MiB)
YOLO26n summary (fused): 122 layers, 2,375,421 parameters, 0 gradients, 5.2 GFLOP
s
val: Fast image access (ping: 0.00.0 ms, read: 573.3183.8 MB/s, size: 26.5 KB)
val: Scanning C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_cla
se3\TareaSemana3-3\valid\labels.cache... 25 images, 0 backgrounds, 0 corrupt: 10
0% ————— 25/25 7.5Mit/s 0.0s

```

	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 50%	all	1/2	3.9s/it 1.2s<3.9s				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	all	2/2	1.5it/s 1.4s	0.883	0.824	0.952	
0.87	Dinosaurio	12	12	0.937	0.833	0.968	
0.889	Llaves	9	9	0.836	0.889	0.975	
0.892	Pin	4	4	0.877	0.75	0.912	
0.829							

```

Speed: 4.8ms preprocess, 10.3ms inference, 0.0ms loss, 0.1ms postprocess per imag
e
Results saved to C:\Users\brand\Documents\GitHub\vision-por-computador\ejercicio_
clase3\runs\detect\val2
Learn more at https://docs.ultralytics.com/modes/val

```

Step 8: Test Your Custom Model on an Image

```

In [13]: # Load your custom trained model
custom_model = YOLO('C:/Users/brand/Documents/GitHub/vision-por-computador/ejerc

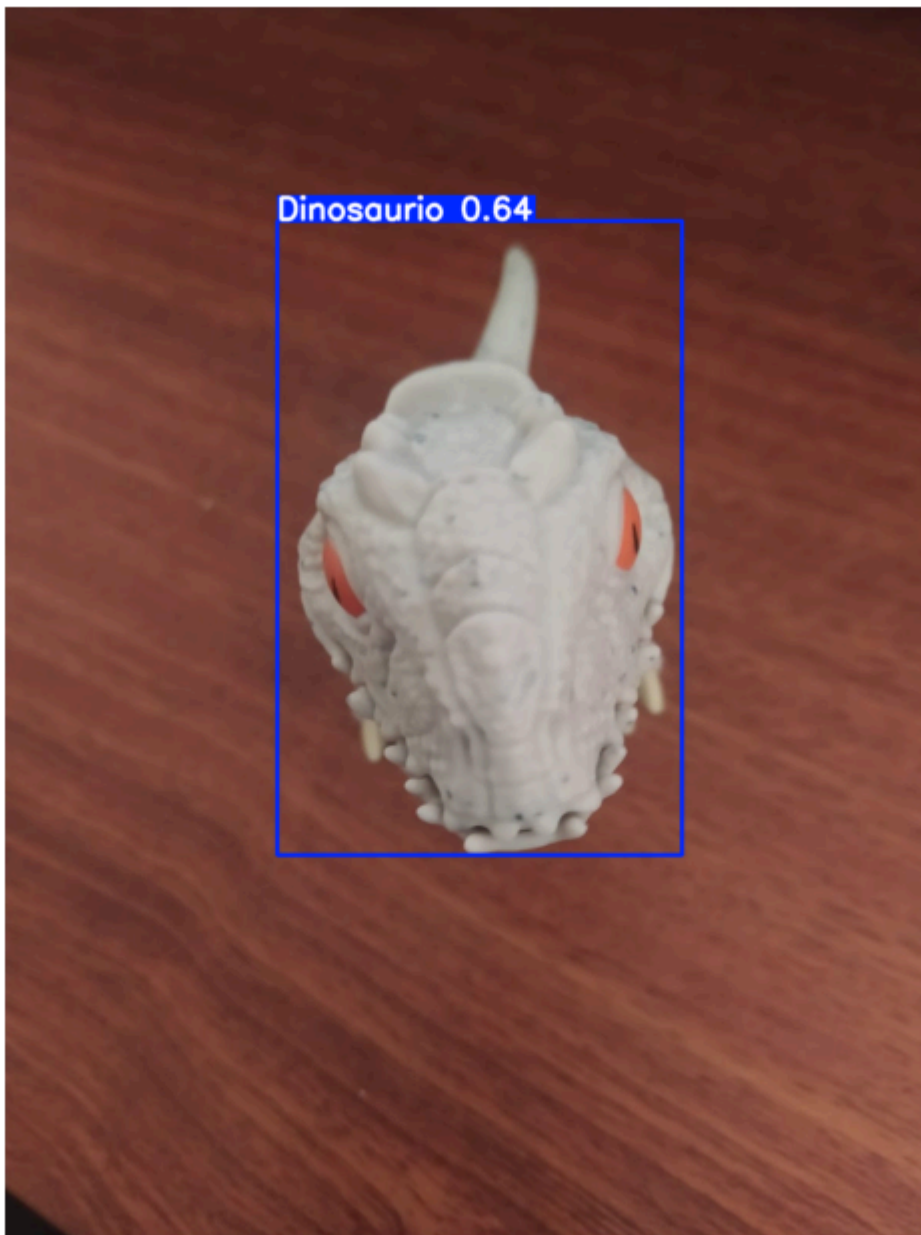
# Test on a sample image
results = model("C:/Users/brand/Downloads/Dinosaurio (8).jpeg", conf=0.25)

```

```
# Display results
%matplotlib inline
for result in results:
    img = result.plot() # Plot detections
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    plt.figure(figsize=(12, 8))
    plt.imshow(img_rgb)
    plt.axis('off')
    plt.show()
```

image 1/1 C:\Users\brand\Downloads\Dinosaurio (8).jpeg: 640x480 1 Dinosaurio, 22.5ms

Speed: 2.7ms preprocess, 22.5ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 480)



Step 9: Test Your Custom Model on Video

```
In [ ]: # Se recargan las librerías necesarias para evitar errores en la captura
import cv2
```

```

from ultralytics import YOLO
from IPython.display import display, clear_output
import matplotlib.pyplot as plt

# Carga del modelo entrenado
model = YOLO("runs/detect/yolo_clase_3/weights/best.pt")

# Inicia la camara
cap = cv2.VideoCapture(0)

try:
    while True:
        ret, frame = cap.read()
        if not ret:
            break

        # Se realiza la prediccion en los frames del video
        results = model(frame)
        annotated_frame = results[0].plot()

        # Se le agrega un label a la deteccion de cada frame
        annotated_frame = cv2.cvtColor(annotated_frame, cv2.COLOR_BGR2RGB)

        # Muestra el frame
        plt.imshow(annotated_frame)
        plt.axis('off')
        clear_output(wait=True)
        display(plt.gcf())

except KeyboardInterrupt:
    # Salir con ctrl c para cerrar la camara
    print("Cámara cerrada.")

finally:
    # Se asegura el cierre de la camara
    cap.release()
    clear_output(wait=True)

```

Cámara cerrada.



Activity: Create Your Custom Detector

1. **Create a RoboFlow account** and set up a new project
2. **Upload at least 25 images** of your custom objects (at least 2)
3. **Annotate all images** with bounding boxes and class labels
4. **Train a YOLO model** on your custom dataset
5. **Test your model** on new images and evaluate its performance
6. **Implement real-time detection** with your custom classes

Grading Rubric:

Criteria	Points	Description
1. Dataset Creation	4	Successfully creates and annotates a custom dataset with at least 25 images and multiple classes in RoboFlow.
2. Model Training	1	Properly trains a YOLOv8 model on the custom dataset with appropriate hyperparameters.
3. Model Evaluation	1	Evaluates model performance with appropriate metrics and provides analysis of results.
4. Custom Detection	3	Implements real-time detection with custom classes and displays results correctly. You have to use video from your webcam, no images or saved videos allowed.
5. Documentation	1	Provides clear documentation of the process, including dataset details and model performance.

Total: 10 points

Tips for Success:

- **Quality over quantity:** Well-annotated images are more important than many poorly annotated ones
- **Class balance:** Try to have a similar number of examples for each class
- **Variety:** Include different angles, lighting conditions, and backgrounds
- **Augmentation:** Use RoboFlow's augmentation features to expand your dataset

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